King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam I Term 222

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. $\int_0^3 \frac{x^2+2}{x+1}\,dx =$

(a) $2 - \ln 4$

(b) $\frac{1}{2} - 2\ln 4$

(c) $3 + 3\ln 4$

(d) $\frac{3}{2} + 3\ln 4$

(e) $\frac{5}{2}$

2. The area of the region in the first quadrant bounded by the graphs of

$$f(x) = x^2 + 2, \qquad g(x) = 4 - x^2, \qquad h(x) = 2 - x$$

is equal to

(a) $\frac{7}{2}$

(b) $\frac{5}{2}$

(c) 4

(d) 3

(e) 2

3. The area of the region bounded by the graph of $f(x) = 4 - x^2$, the x-axis and the vertical lines $x = 1$ and $x = 2$ is given by (choose c_i to be the right endpoint)

(a) $\lim_{n\to\infty} \left(\frac{4}{n}\sum_{i=1}^n 1 - \frac{1}{n^2}\sum_{i=1}^n i - \frac{2}{n^3}\sum_{i=1}^n i^2\right)$

(b) $\lim_{n\to\infty} \left(\frac{1}{n}\sum_{i=1}^n 2 - \frac{3}{n^2}\sum_{i=1}^n i + \frac{1}{n^3}\sum_{i=1}^n i^2\right)$

(c) $\lim_{n\to\infty} \left(\frac{1}{n}\sum_{i=1}^n 3 - \frac{1}{n^2}\sum_{i=1}^n i\right)$

(d) $\lim_{n\to\infty} \left(\frac{3}{n}\sum_{i=1}^n 1 - \frac{2}{n^2}\sum_{i=1}^n i - \frac{1}{n^3}\sum_{i=1}^n i^2\right)$

(e) $\lim_{n\to\infty} \left(\frac{1}{n}\sum_{i=1}^n 3 + \frac{1}{n^2}\sum_{i=1}^n i + \frac{1}{n^3}\sum_{i=1}^n i^2\right)$

4. The average value of $f(x) = \sqrt{x}$ over the interval $[1, 4]$ is

(a) $\frac{7}{3}$

(b) $\frac{14}{9}$

(c) $\frac{7}{9}$

(d) $\frac{2}{9}$

(e) $\frac{8}{3}$

5. If $F(x) = \int_3^{x^3} \frac{3}{t^3+3}\, dt$, then $F'(x) =$

(a) $\frac{9x^2}{x^9+3}$

(b) $\frac{9x^2}{x^6+3}$

(c) $\frac{-9x^2}{x^9+3}$

(d) $\frac{3}{x^9+3} - \frac{1}{10}$

(e) $\frac{3}{x^9+3}$

6. The area of the region bounded by the curves $y = (x+1)^2$ and $y = 1 - |x|$ is equal to

(a) $\frac{1}{6}$

(b) $\frac{1}{3}$

(c) $\frac{2}{3}$

(d) $\frac{5}{6}$

(e) 1

7. If

$$\int_{-1}^{1} f(x)\,dx = -3 \quad \text{and} \quad \int_{0}^{1} f(x)\,dx = 5,$$

then $\int_{-1}^{0} f(x)\,dx =$

(a) -8

(b) $-\dfrac{3}{2}$

(c) 8

(d) 5

(e) 2

8. $\int_{e}^{e^2} \frac{1}{x(\ln x^2)^3}\,dx =$

(a) $\dfrac{5}{16}$

(b) $\dfrac{3}{64}$

(c) $\dfrac{1}{64}$

(d) $\dfrac{5}{32}$

(e) $\dfrac{3}{16}$

9. The velocity (in m/s) of a particle moving along a line is given by

$$v(t) = t^2 - t - 12.$$

The total distance the particle travels over the interval $0 \le t \le 5$ is

(a) $\dfrac{77}{2}$

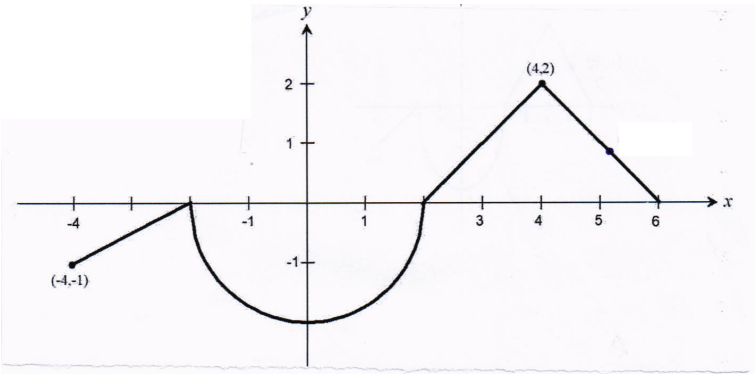
(b) $\dfrac{125}{6}$

(c) $\dfrac{53}{2}$

(d) $\dfrac{55}{3}$

(e) $\dfrac{67}{3}$

10. $\int_{-4}^{5} f(x)\,dx =$ (the graph of $y = f(x)$ is given below)



(a) $3 - 2\pi$

(b) $\dfrac{5}{2} - 2\pi$

(c) $5 + 2\pi$

(d) $\dfrac{3}{2} - 2\pi$

(e) $6 + 2\pi$

11. Using the Midpoint Rule with $n = 3$, the area of the region bounded by the graph of $f(x) = \sin^2 x$ and the x-axis over the interval $[0, \pi]$ is approximately equal to

(a) $\dfrac{2\pi}{3}$

(b) $\dfrac{\pi}{3}$

(c) $\dfrac{\pi}{2}$

(d) $\dfrac{\pi}{\sqrt{3}}$

(e) $\dfrac{\pi}{4}$

12. If $f(x) = \int_{\sqrt{x}}^{x} e^{t^2}\,dt$, then $f'(4) =$

(a) $\dfrac{e^4}{4} - e^{16}$

(b) $4e^{16} - e^4$

(c) $8e^{16} - 4e^4$

(d) $e^{16} - e^4$

(e) $e^{16} - \dfrac{e^4}{4}$

13. $\sum_{i=1}^{n} \frac{8i+3}{n^2} =$

(a) $4 + \dfrac{3}{n^2}$

(b) $4 + \dfrac{2}{n} + \dfrac{3}{n^2}$

(c) $4 + \dfrac{7}{n}$

(d) $4 + \dfrac{4}{n}$

(e) $4 + \dfrac{3}{n}$

14. $\int t \sec t^2 \cdot \left(2 \sec t^2 + \tan t^2\right) dt =$

(a) $\dfrac{1}{2} \tan t^2 + \dfrac{1}{2} \sec t^2 + C$

(b) $\ln \left|\sec t^2 + \tan t^2\right| + C$

(c) $\tan t^2 + \dfrac{1}{2} \sec t^2 + C$

(d) $\tan^2 t^2 + \sec^2 t^2 + C$

(e) $\sec t^2 \cdot \tan t^2 + C$

15. $\int \frac{2x-1}{\sqrt{x+3}}\, dx =$

(a) $(2x - 15)\sqrt{(x+3)^3} + C$

(b) $\dfrac{1}{3}(x+3)(2x-1) + C$

(c) $2(2x - 15)\sqrt{x+3} + C$

(d) $\dfrac{2}{3}(2x-1)\sqrt{x+3} + C$

(e) $\dfrac{2}{3}(2x-15)\sqrt{x+3} + C$

16. If f is an even function and $\int_0^1 f(x)\, dx = 5$, then $\int_{-1}^1 (3x-4) f(x)\, dx =$

(a) 20

(b) -40

(c) -20

(d) 40

(e) 0

17. $\int_0^{\pi/6} \frac{4\cos x}{1+4\sin^2 x}\,dx =$

(a) π

(b) $\dfrac{\pi}{4}$

(c) $\dfrac{\pi}{2}$

(d) 2π

(e) 4π

18. $\int_0^{\ln 2} \tanh x\,dx =$

(a) $\ln\left(\dfrac{3}{2}\right)$

(b) $\ln(2)$

(c) $\ln\left(\dfrac{5}{4}\right)$

(d) 1

(e) 0

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$x-1+\frac{3}{x+1}$: $\left[\frac{x^2}{2}-x+3\ln\lvert x+1\rvert\right]_0^3=\frac{3}{2}+3\ln 4$.
2	**(e)**	$\int_0^1(x^2+x)\,dx+\int_1^2(2+x-x^2)\,dx=\frac{5}{6}+\frac{7}{6}=2$.
3	**(d)**	$c_i=1+\frac{i}{n}$: $f(c_i)\Delta x=\frac{3}{n}-\frac{2i}{n^2}-\frac{i^2}{n^3}$.
4	**(b)**	$\frac{1}{3}\int_1^4\sqrt{x}\,dx=\frac{2}{9}(8-1)=\frac{14}{9}$.
5	**(a)**	$F'(x)=\frac{3}{(x^3)^3+3}\cdot 3x^2=\frac{9x^2}{x^9+3}$.
6	**(a)**	Intersect at $x=-1,0$: $\int_{-1}^0(-x^2-x)\,dx=\frac{1}{6}$.
7	**(a)**	$-3=\int_{-1}^0 f+5$, so $\int_{-1}^0 f=-8$.
8	**(b)**	$u=\ln x^2$: $\frac{1}{2}\int_2^4 u^{-3}du=\frac{3}{64}$.
9	**(a)**	$\lvert v\rvert$ split at $t=4$: $\frac{104}{3}+\frac{23}{6}=\frac{77}{2}$.
10	**(b)**	$-1-2\pi+\left(4-\frac{1}{2}\right)=\frac{5}{2}-2\pi$.
11	**(c)**	Midpoints $\frac{\pi}{6},\frac{\pi}{2},\frac{5\pi}{6}$: $\frac{\pi}{3}\left(\frac{1}{4}+1+\frac{1}{4}\right)=\frac{\pi}{2}$.
12	**(e)**	$f'(x)=e^{x^2}-\frac{e^x}{2\sqrt{x}}$, so $f'(4)=e^{16}-\frac{e^4}{4}$.
13	**(c)**	$\frac{1}{n^2}\left(4n(n+1)+3n\right)=4+\frac{7}{n}$.
14	**(c)**	$u=t^2$: $\frac{1}{2}\int(2\sec^2 u+\sec u\tan u)\,du=\tan t^2+\frac{1}{2}\sec t^2+C$.
15	**(e)**	$u=x+3$: $\frac{4}{3}u^{3/2}-14u^{1/2}+C=\frac{2}{3}(2x-15)\sqrt{x+3}+C$.
16	**(b)**	$xf(x)$ odd gives 0; $-4\cdot 2\cdot 5=-40$.
17	**(c)**	$u=2\sin x$: $\int_0^1\frac{2\,du}{1+u^2}=2\cdot\frac{\pi}{4}=\frac{\pi}{2}$.
18	**(c)**	$u=\cosh x$: $\int_1^{5/4}\frac{du}{u}=\ln\frac{5}{4}$.